

Traffic Crash Analytics & Safety Intelligence Platform

Transportation Analytics

CH.SAI SIDDARTHA REDDY

Under the Guidance of
SHADIYA P P

Abstract

- The Traffic Crash Analytics & Safety Intelligence Platform is a data analytics project developed using SQL, Python, SQLite, and Streamlit to analyze traffic crash data and generate meaningful insights related to road safety, crash trends, injury patterns, and contributing risk factors. The project applies advanced SQL techniques such as aggregations, joins, CTEs, and window functions to identify high-risk locations, peak crash timings, and major causes of accidents. An interactive Streamlit dashboard is used to display analytical results and business insights, helping support data-driven decision making for traffic management, emergency response, and urban safety planning.

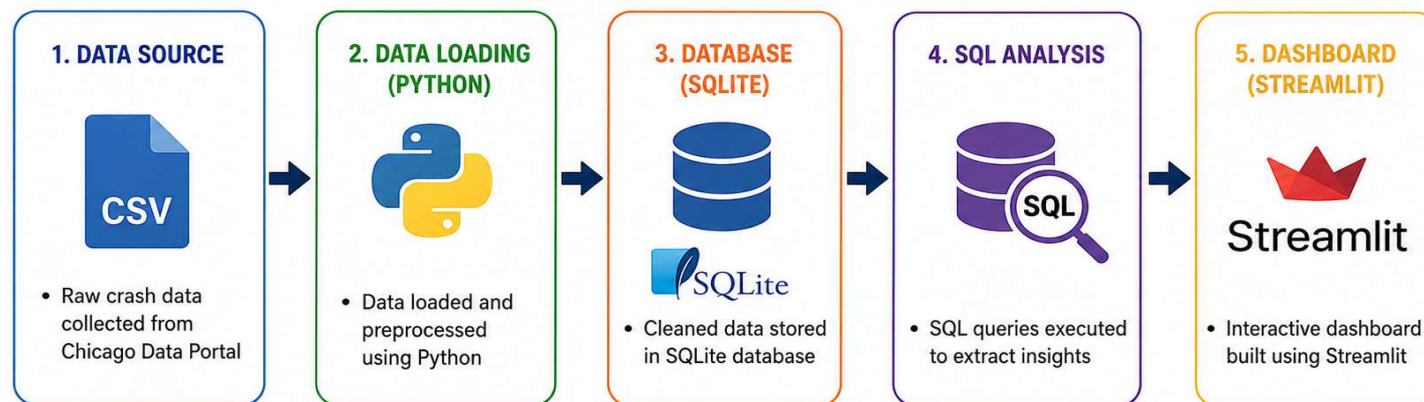
Problem Statement

- Traffic crash data is large and complex, making manual analysis difficult.
- Identifying high-risk locations, crash patterns, and contributing factors requires advanced analytical techniques.
- A data-driven analytical platform is needed to generate meaningful insights for improving road safety and decision making.

Dataset Overview

Property	Details
Dataset Name	Chicago Traffic Crashes Dataset
Source	Chicago Data Portal — data.cityofchicago.org
Format	CSV (Comma-Separated Values)
Total Records	600,000+ Rows
Time Range	2015 to Present
Key Attributes	Crash Type, Weather Condition, Injury Details, Street Name, Crash Time, Location
Dataset Status	Pre-cleaned and Structured for Analysis

System Workflow / Architecture

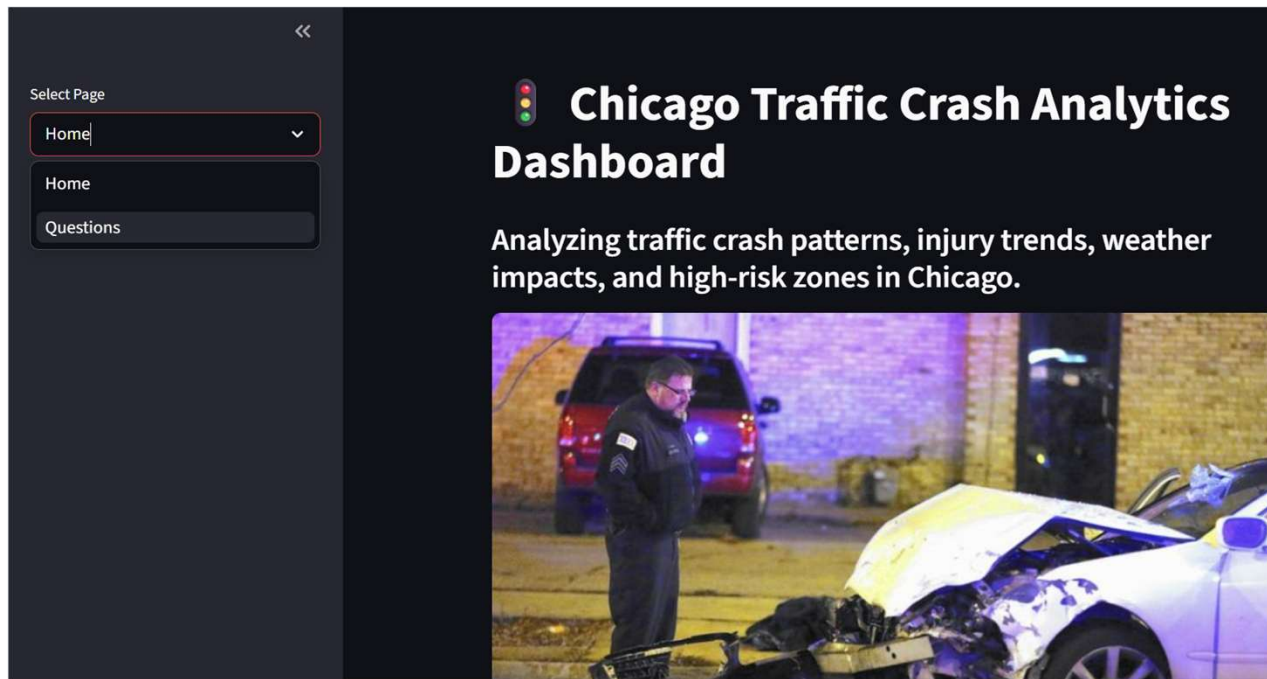


Key Analytical Queries

Present major analyses performed:

- Top dangerous weather conditions
- Streets with highest injury crashes
- Peak crash hours
- High-risk time slots
- Crash growth trends
- Top contributing causes
- Hotspot zone identification

Dashboard Screenshot



Challenges Faced

- ▶ Understanding the structure of a large dataset containing over 600,000 crash records.
- ▶ Connecting the SQLite database with Python using SQLAlchemy.

Conclusion

- ▶ This project analyzed Chicago Traffic Crash data using SQL, Python, Pandas, SQLite, and Streamlit to identify crash patterns, injury trends, high-risk locations, and contributing factors. The insights generated through data analysis and visualization can help support informed traffic safety decisions.

Thank You